

YUNCHENG, China

WMO: 539590

Lat: 35.1118N

Lon: 111.0670E

Elev: 377

StdP: 96.88

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.7	-6.8	-22.6	0.5	-1.1	-19.7	0.7	-0.6	8.4	2.2	7.3	2.6	1.4	20	0.424

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	36.5	22.3	35.2	22.3	33.8	22.0	25.7	32.4	25.1	31.7	24.4	30.7	3.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.7	29.7	23.2	18.8	29.0	22.4	17.9	28.5	82.3	32.6	79.2	31.7	76.3	30.9	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	7.0	5.9	-11.5	39.2	1.8	1.1	-12.8	40.0	-13.9	40.7	-14.9	41.3	-16.3	42.1		
			-12.7	26.6	1.8	0.9	-14.0	27.2	-15.1	27.7	-16.1	28.2	-17.4	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.0	-0.2	4.0	10.4	16.7	21.8	26.3	28.1	26.6	21.5	15.2	7.6	1.3
	DBStd	10.35	2.89	3.76	4.31	4.12	3.40	3.17	2.95	3.25	3.60	3.51	3.67	2.85
	HDD10.0	898	316	169	48	3	0	0	0	0	0	3	89	269
	HDD18.3	2278	574	401	248	77	9	1	0	0	10	108	323	528
	CDD10.0	2721	0	2	60	204	366	488	559	514	345	164	17	0
	CDD18.3	1058	0	0	2	28	116	239	301	256	105	11	0	0
	CDH23.3	11474	0	0	28	308	1166	2738	3622	2721	811	82	0	0
	CDH26.7	5199	0	0	4	87	451	1343	1818	1219	265	13	0	0
Wind	WSAvg	2.4	2.0	2.5	2.7	2.7	2.7	2.6	2.6	2.3	2.0	1.9	2.0	
Precipitation	PrecAvg	507	4	9	17	36	53	56	98	79	81	47	23	4
	PrecMax	843	14	19	44	112	140	123	268	183	280	123	86	15
	PrecMin	305	0	0	0	2	9	18	29	8	14	5	1	0
	PrecStd	113	4	6	13	23	32	28	52	43	59	31	22	5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.9	17.7	26.4	32.4	35.4	38.0	38.6	37.2	34.8	28.3	20.3	11.7
		MCWB	3.4	8.2	13.6	17.4	18.0	21.0	23.7	23.8	22.1	16.3	11.2	4.5
	2%	DB	8.8	14.9	23.2	29.1	32.9	36.2	36.6	35.4	31.7	25.6	17.9	9.8
		MCWB	2.4	6.7	11.8	16.1	18.0	20.4	23.5	23.4	20.8	16.0	10.3	3.7
	5%	DB	7.2	12.8	20.6	26.8	31.1	34.7	35.1	33.8	29.6	23.4	16.0	8.2
		MCWB	1.5	5.5	10.6	15.0	17.3	20.4	23.7	23.3	19.9	15.2	9.7	2.7
	10%	DB	5.6	10.9	18.2	24.6	29.3	33.0	33.7	32.1	27.7	21.5	14.0	6.7
		MCWB	0.6	4.6	9.4	14.2	17.0	20.1	23.5	23.1	19.3	14.5	8.9	2.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.3	9.4	15.2	20.2	21.1	24.8	27.0	26.3	24.9	19.0	13.0	6.5
		MCDB	8.8	16.3	22.7	27.5	29.2	32.5	34.1	32.4	31.5	23.9	17.0	9.0
	2%	WB	3.1	7.5	12.9	17.8	20.0	23.5	26.1	25.5	22.8	17.6	11.7	4.5
		MCDB	7.5	13.1	21.4	25.5	28.0	31.1	32.8	31.8	29.0	22.2	15.8	8.2
	5%	WB	2.0	6.4	11.4	16.4	19.3	22.5	25.4	24.8	21.6	16.7	10.6	3.5
		MCDB	6.7	11.6	18.9	24.5	27.0	30.1	32.1	31.0	26.7	21.0	14.5	7.0
	10%	WB	0.9	5.2	10.1	15.2	18.5	21.7	24.7	24.2	20.5	15.7	9.5	2.5
		MCDB	5.2	10.2	17.1	22.5	26.1	29.1	31.4	30.2	25.5	19.8	13.3	6.1
Mean Daily Temperature Range	5% DB	MDBR	9.5	10.0	10.8	11.3	11.4	10.6	8.7	8.3	8.7	9.2	9.4	9.2
		MCDBR	12.4	13.6	15.1	15.3	15.4	13.7	10.9	10.3	11.7	12.8	12.4	11.6
		MCWBR	7.4	7.3	6.9	5.9	4.9	3.9	3.0	2.8	4.1	5.5	6.4	6.8
	5% WB	MCDDBR	11.1	11.9	13.2	12.3	11.3	10.3	9.0	8.5	9.1	9.7	10.0	9.8
		MCWBR	6.9	6.7	6.4	5.6	4.5	3.7	3.1	2.9	3.9	4.7	5.7	6.1
Clear-Sky Solar Irradiance	taub	0.518	0.606	0.616	0.623	0.623	0.696	0.710	0.735	0.666	0.647	0.564	0.493	
	taud	1.739	1.601	1.613	1.636	1.689	1.592	1.615	1.550	1.643	1.617	1.708	1.806	
	Ebn at Noon	628	619	667	695	704	653	640	610	624	579	580	621	
	Edh at Noon	180	231	247	252	242	267	259	272	237	224	184	159	
All-Sky Solar Radiation	RadAvg	2.55	3.06	4.18	5.01	5.46	5.45	4.96	4.47	3.63	3.17	2.62	2.39	
	RadStd	0.28	0.52	0.43	0.43	0.42	0.49	0.54	0.55	0.37	0.47	0.38	0.30	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% DP	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	-0.83	-1.76	N/A	-0.77	-0.90	N/A	N/A	N/A	N/A	
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page